

Mrinal Mohit

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TECHNICAL SKILLS

Languages: Python, Golang, Java, C++, SQL, JavaScript, Dart, Embedded C

AI & Agentic Systems: LLM APIs, Prompt Engineering, Multi-Agent Systems, LangGraph, Tool Use & Function Calling, Structured Outputs, Guardrails & Validation, Local LLMs (Ollama)

Frameworks/Libraries: FastAPI, Spring Boot, Flask, React, Flutter

Databases & Caching: PostgreSQL, MongoDB Atlas, MySQL, Firestore, Redis

Messaging Systems: RabbitMQ (AMQP), MQTT

Cloud & DevOps: AWS (Lambda), GCP, Firebase, Docker, Jenkins, CI/CD

Systems & Architecture: Distributed Systems, Event-Driven Architecture, Async Programming, Microservices, Rate Limiting, Circuit Breakers

Developer Tools: GitHub, GitHub Copilot, Maven, Postman, Figma

Hardware/IoT: ESP32, GPS Module, Sensors, Arduino

PROJECTS

Project Synapse – Autonomous AI Agent for Last-Mile Coordination

Tech Stack: Python – Ollama (LLaMA3) – AWS Lambda – API Gateway – Docker

- Developed an **autonomous LLM-powered agent** for real-time resolution of last-mile delivery disruptions, coordinating between customers, drivers, and merchants.
- Architected a **5-layer modular system** with tool-based execution across **20+ logistics APIs**, enabling multi-step planning and execution with **80% success rate**.
- Implemented **iterative reasoning loops** (CoT + tool-use) with structured action parsing, reducing resolution time to **10–60 seconds** vs hours manually.
- Deployed on **AWS Lambda** with API Gateway integration, enabling scalable serverless execution and real-time decision-making pipelines.

MeetFlow AI – Multi-Agent Enterprise Workflow Automation System (GitHub)

Tech Stack: Python – FastAPI – LangGraph – Gemini 2.0 Flash – PostgreSQL – Redis – Docker

- Designed a **7-agent LangGraph-based orchestration system** to automate enterprise workflows (Jira, Slack, HR) with **fault-tolerant checkpointing** and audit trails.
- Engineered **production-grade reliability** using circuit breakers, retry logic, rate limiting, and Redis caching, achieving **2–3× throughput improvement**.
- Built **LLM-powered decision extraction and verification pipeline**, enabling autonomous task execution with **human-in-the-loop approval** for high-risk actions.
- Implemented async FastAPI backend with connection pooling and priority queues, supporting **20–30 req/s** and reducing response latency by **30–50%**.

Clara Answers Pipeline – AI Voice Agent Onboarding Automation

Tech Stack: Python – Ollama (LLaMA3) – Pydantic – Docker – n8n

- Built a **two-stage LLM pipeline** to extract and validate structured business data from call transcripts, reducing manual configuration time from **hours to minutes**.
- Engineered an **anti-hallucination validation system** using Pydantic schemas and rule-based checks, achieving **95%+ detection accuracy** of fabricated outputs.
- Designed a **modular, idempotent pipeline** with retry logic and batch processing, enabling scalable automation at **zero API cost** using local LLM inference.
- Integrated webhook-based automation with n8n and implemented structured logging, ensuring **traceability and auditability** across pipeline stages.

GPS-Based Dynamic Toll Collection System (GitHub)

Tech Stack: Spring Boot – Embedded C – PostgreSQL – Redis – RabbitMQ – Docker – Kubernetes

- Engineered **Embedded C** firmware for **ESP32-WROOM-32** with **NEO-6M GPS** and **SIM800L GSM** modules; implemented **MQTT** client with QoS 1 for reliable telemetry transmission, achieving **3m GPS accuracy** and **99%** message delivery rate
- Built **Spring Boot application** processing real-time GPS data from **ESP32** devices; implemented **PostGIS geofencing** and dynamic pricing with traffic/weather-based multipliers.
- Engineered **JWT authentication**, **Redis**-based rate limiting, and fraud detection achieving **95% accuracy**; deployed event-driven architecture using **RabbitMQ** for async processing.
- Containerized with **Docker/Kubernetes** achieving **98% uptime** and **<200ms** response time; integrated **Grafana** monitoring with zero-downtime deployments.

EXPERIENCE

Software Developer Intern

Mar 2025 – Apr 2025

Know Your Trips, Dublin

- Led a **team of 5 developers**, conducted **15+ code reviews**, and **mentored** team members to deliver **scalable mobile features**, improving sprint velocity by **15%**.
- Integrated **JWT authentication** and **Firebase Cloud Messaging**, strengthening **system reliability** by **30%**.
- **Automated** deployments through **CI/CD**, reducing release cycles by **20%**.
- Contributed to **QA testing** and debugging, enhancing app stability by **25%**.

Tech Stack: Flutter, Dart, GetX, MongoDB, AWS, and Firebase.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (*Credential*)
- Oracle Certified Java Professional (Java SE 17) (*Credential*)
- IBM Introduction to Cloud (*Credential*)

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

CGPA: 8.21/10

Expected May 2027

Co-CURRICULAR ACTIVITIES

- Max. rated 1188 on Codeforces.
- Participated in hackathons focusing on fintech automation and cloud-based scalability.